Communication and data transmission MID exam .

1) The interface circuit between your personal computer & wired telephone network is :

a. Antenna
b. Codec
c. Modem
d. Fiber

2) DSL stands for

a. Digital single line
b. Digital system line
c. Digital submission line
d. Digital subscriber line

3) the most difficult medium to tape is :

a. Wireless medium
b. Fiber
c. Coax cable
d. Twisted pair

4) given a channel with capacity of 20 Mbps & Bandwidth of 3MHz . Assuming white noise , what SNR is required to achieve the capacity ?

a. 101 dB
b. 102 dB
c. 100 dB
d. 103 dB

5)if an amplifier has 30 dB voltage gain . what voltage ratio does the gain represent

a. 31.6 V
b. 31.6
c. 31.6 mV
d. 3.16

6) The power line voltage in Jordan can be expressed as:

a. $230 \sin(2\pi * 50t)$
b. $170 \sin(2\pi * 60t)$
c. $230 \sin(2\pi * 60t)$
d. $230 \sin(2\pi * 360t)$

7) The most of energy in the signal is contained in :

a. Spectrum
b. Last harmonic
c. Bandwidth
d. Harmonics

8) Radio signals are made up to

a. Voltages and currents
b. Electrons and protons
c. Noise and data
d. Electric and magnetic fields

9) The communications medium causes the signal to be :

a. Interfered with
b. Attenuated
c. Amplified
d. Modulated

10) if signal has DC component , then :

a. It has f=0 & a non-zero average amplitude
b. It has f≠0 & a non-zero average amplitude
c. It has f≠0 & a zero average amplitude
d. It has f=0 & a zero average amplitude

11) Which of the following is correct:

a. The higher data rate of signal , the greater is the bandwidth & the greater the distortion
b. The higher data rate of signal , the greater is the bandwidth & the smaller the distortion
c. The higher data rate of signal , the smaller is the bandwidth & the greater the distortion
d. The smaller data rate of signal, the greater is the bandwidth & the greater the distortion

12) delay distortion is caused by :

a. Different velocity of propagation for different harmonics
b. Constant velocity of propagation for different harmonics
c. Different velocity of propagation in free space for different harmonics
d. Constant velocity of propagation in free space for different harmonics

13) the following medium is limited in distance , bandwidth & data rate

a. Coax cable
b. Waveguide
c. Twisted pair
d. Free space

14)the following mode is used in telephony

a. Step index multimode
b. Graded index multimode
c. Single index multimode
d. Single mode propagation

15) usually human vision allocated with respect to wavelength in

a. Megameters
b. Nanometers
c. Millimeters
d. Micrometers

16) Radio waves travel more slowly near the ground , because :

a. The refractive index of atmosphere decreases with height
b. The refractive index of atmosphere increases with height
c. The refractive index of atmosphere the same regardless to the height
d. The reflection index of atmosphere decreases with height

17) The combination of Tx/Rx in the satellite is called :

a. Uplink
b. Transponder
c. Downlink
d. Repeater

18) multipath could be :

a. Could be advantage and could be disadvantage
b. Always advantage
c. Always disadvantage
d. Always a merit

19) international audio broadcasting is usually used :

a. Ground wave propagation
b. Direct wave propagation
c. Sky wave propagation
d. Surface wave propagation

20) typical speech has a dynamic range of about :

a. 2.5 dB
b. 250 dB
c. 0.25 Db
d. 25 dB

Answers:-

1	2	3	4	5	6	7	8	9	10
C	**D**	**B**	**A**	**B**	**A**	**C**	**D**	**B**	**A**
11	12	13	14	15	16	17	18	19	20
B	**A**	**C**	**D**	**B**	**A**	**B**	**A**	**C**	**D**